[Project Title: a specific, question-style title]

[Optional subtitle: data and method in one line]

Member	UID	Member	UID
[Name 1]	[UID]	[Name 4]	[UID]
[Name 2]	[UID]	[Name 5 (if any)]	[UID]
[Name 3]	[UID]		

Team: [Team Name] **Date:** [DD Nov 2026]

Required on the title page: course name, project title, student names & UIDs, team name, date. Replace blue [placeholders]; delete all grey guidance text before exporting. Max 10 pages including this one.

[Action title: why this problem matters, in one sentence]

Speaker: [Name] · ~0:45

Who is affected, what decision the results inform, why current knowledge is not enough. Then the research questions you will answer.

Motivation

- [Context of the problem]
- [Who would use the result, and for what decision]
- [Gap: what is unknown or done poorly today]

RQ1

[A specific question answerable with the data]

RQ2

[A specific question answerable with the data]

RQ3

[A specific question answerable with the data (optional)]

[Action title: e.g. N listings with p features after cleaning]

Speaker: [Name] · ~0:45

Source and link, snapshot date / time span, unit of observation, response variable, main predictor groups. Show every cleaning rule and how many rows it removed.

- **Source:** [dataset, provider, URL, access date]
- **Unit:** [each row is ...]
- **Response:** [variable; transformation, e.g. log]
- **Predictors:** [groups + engineered features]
- **Known issues:** [missingness, imbalance, leakage risks]

Cleaning step	Rows left
[Raw data]	[N0]
[Remove duplicates / inactive]	[N1]
[Drop missing response]	[N2]
[Outlier rule, e.g. top 1%]	[N3]
Final sample	[N]

[Action title: the EDA finding that shaped the modelling]

Speaker: [Name] · ~0:45

Two or three figures only, each leading to a modelling decision (e.g. skewed response -> log; non-linear trend -> splines; strong spatial pattern -> location features).

Figure: e.g. distribution of the response

What it shows -> the decision it led to

Figure: e.g. key relationship or map

What it shows -> the decision it led to

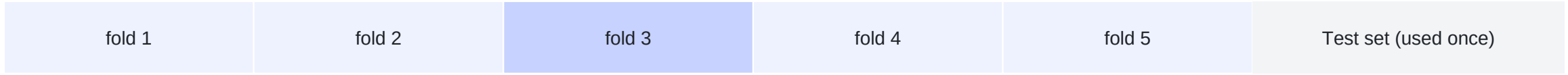
[Action title: a model ladder judged on unseen data]

Speaker: [Name] · ~0:50

Each step must earn its place against the previous one. Show how data was split and how hyper-parameters were tuned. The test set is used once, at the end.

increasing flexibility → decreasing interpretability

Validation design



Training set ([80%]): [k]-fold CV for tuning · seed [xxxx] · metric [RMSE / AUC]

[Action title: what the interpretable models already tell us]

Speaker: [Name] · ~0:40

Baseline vs linear vs regularised. Show the coefficient path or the top coefficients with signs, and what they mean in plain words.

Figure: e.g. Lasso coefficient path / top coefficients

KEY TAKEAWAY

[One-sentence finding with a number, e.g. Lasso keeps 18 of 60 features; X raises the response by about 12%]

- [Supporting point 1]
- [Supporting point 2]

[Action title: which model wins, and by how much]

Speaker: [Name] · ~0:55

Same split and folds for every model. Report CV and test metrics with 95% bootstrap intervals. If intervals overlap, say the models are comparable.

Model	Key hyper-parameters	CV metric (mean ± sd)	Test metric (95% CI)	Fit time
Baseline	—	[x]	[x [a, b]]	[s]
OLS / Logistic	—	[x]	[x [a, b]]	[s]
Ridge / Lasso	[lambda]	[x]	[x [a, b]]	[s]
GAM	[smoothing]	[x]	[x [a, b]]	[s]
Random forest	[trees, mtry]	[x]	[x [a, b]]	[s]
Gradient boosting	[depth, rate]	[x]	[x [a, b]]	[s]

KEY TAKEAWAY

[e.g. Boosting cuts test RMSE by 18% vs OLS, but only 3% vs GAM, so the GAM is preferred for interpretability]

[Action title: the main drivers, and how they act]

Speaker: [Name] · ~0:55

Variable importance answers 'what matters'; partial dependence / GAM smooths answer 'how'. Translate each pattern into plain language. Avoid causal wording.

Figure: permutation importance (top 10)

Figure: partial dependence / smooth term

KEY TAKEAWAY

[e.g. Location explains most of the variation; the effect of distance flattens beyond 3 km]

[Action title: the findings hold under these checks]

Speaker: [Name] · ~0:40

Show that conclusions do not depend on one arbitrary choice. Include at least one sub-group comparison and one alternative cleaning rule.

Check	What changed	Result	Conclusion still holds?
[Sub-sample A vs B]	[e.g. split by group]	[metric / effect in each]	[Yes / Partly / No]
[Alternative outlier rule]	[e.g. 1% -> 5% trimming]	[metric]	[Yes / Partly / No]
[Drop strongest feature]	[feature name]	[metric]	[Yes / Partly / No]
[Different seed / split]	[5 repeats]	[mean ± sd]	[Yes / Partly / No]

[One or two sentences on any check that changed the story, and why.]

[Action title: the answer, in one sentence]

Speaker: [Name] · ~0:50

Answer each RQ directly with a number. Give 2 to 3 recommendations for the users named on page 2. Be honest about limitations.

RQ1: [direct answer + key number]

RQ2: [direct answer + key number]

RQ3: [direct answer + key number]

Recommendations

- [For [user]: action 1]
- [For [user]: action 2]

Limitations

- [e.g. observational data: association, not causation]
- [e.g. single snapshot / time span]
- [e.g. unobserved variables]

Future work

[one realistic next step]

Source code for every result: **Team-[XX]-Appendix.pdf**